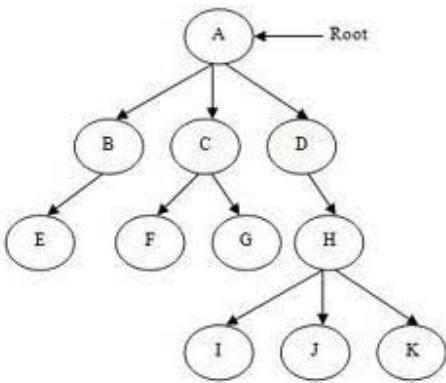


Total No of Questions: [8]		SEAT NO. :	
[Total No. of Pages :3]			
S.E. (Computer) 2008 Course			
Data Structure and Problem Solving			
Time: 2 Hours		Max. Marks : 50	
Instructions to the candidates:			
1) Answers Q.1 or Q.2,Q.3 or Q.4,Q.5 or Q.6,Q.7 or Q.8.			
2) Neat diagrams must be drawn wherever necessary.			
3) Figures to the right side indicate full marks.			
4) Assume Suitable data if necessary			
SECTION I			
Q1)	a)	Find the frequency count for the following code. for(i=1;i<=n;i++) { for(i=1;i<=n;i++) { for(i=1;i<=n;i++) { sum=sum+I; } } }	[3]
	b)	Write a function for selection sort using templates.	[3]
	c)	What is ADT? Write an ADT for Deques.	[6]
OR			
Q2)	a)	Explain different Asymptotic notations.	[3]
	b)	Convert the following tree into Binary tree 	[6]
	c)	Consider the following tree given in the problem. Show a Postorder, Preorder and In order Traversal of the tree	[3]

Q3)	a)	Sort the digraph for topological sort refer figure 1 below FIGURE 1	[3]
	b)	Convert given graph into MST refer figure 2 below. FIGURE 2	[3]
	c)	What is collision? What Are different collision resolution techniques?	[3]
	d)	Draw a binary search tree for the following data 10, 08, 15, 12, 13, 07, 09, 17, 20, 18, 04, 05	[3]
		OR	
Q4)	a)	Explain with suitable example the various storage structures for the graph	[6]
	b)	Construct the AVL tree for the following data by inserting each data item one at a time. 15, 20, 24, 10, 13, 7, 30, 36, 25	[6]
Q5)	a)	Sort the following data in ascending order using heap sort. 15, 19, 10, 7, 17, 16	[6]
	b)	Write an algorithm to search an element in a B Tree	[4]
	c)	Define sequential file organization and state its advantages and disadvantages	[3]
		OR	
Q6)	a)	What is ISMA in file organization? Explain Advantages & Disadvantages of	[6]

		sequential file organization.	
	b)	What is a B+ tree? Give structure of its internal node. What are the order of B+ tree & Characteristics of B+ tree.	[7]
Q7)	a)	Explain in detail the models used for parallel computation	[6]
	b)	Write a parallel algorithm to perform the addition of the given numbers using complete binary tree method	[7]
		OR	
Q8)	a)	Write a parallel algorithm for pointer doubling. Explain with suitable example	[7]
	b)	Write a parallel algorithm for odd-even merge sort.	[6]